THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 6 — MATHS · HARD LEVEL CHAPTER TEST SET 3

Chapter 2: **Lines and Angles** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The hour hand of a clock turns through how many degrees in 50 minutes? 1

(a) 30 **(b)** 24 **(c)** 26 **(d)** 25

2. 6 lines are drawn through a point O. The number of straight angles at O (pairs of opposite rays) is: 1

(a) 5 **(b)** 18 **(c)** 7 **(d)** 6

3. Rays OA, OB, OC, OD make equal angles around O (a full turn). The measure of each angle is: 1

(a) 45 **(b)** 55 **(c)** 35 **(d)** 135

4. A wheel of 8 equal spokes: the angle between two neighbouring spokes is 45°. The angle between spoke 1 and spoke 4 (going one way) is: 1

(a) 135 **(b)** 125 **(c)** 145 **(d)** 165

5. At 8:10 the smaller angle between the clock hands is: 1

(a) 180 **(b)** 175 **(c)** 190 **(d)** 170

6. A wheel turns 9 full turns and then a further quarter turn. The total angle in degrees is: 1

(a) 3240 **(b)** 3330 **(c)** 3420 **(d)** 3510

7. A spinner starts pointing North and turns clockwise 180° then anticlockwise 45°. The final angle clockwise from North is: 1

(a) 135 **(b)** 225 **(c)** 125 **(d)** 145

8. The smaller angle between the clock hands at 5 o'clock is 150°. The smaller angle between the hands at 7 o'clock is: 1

(a) 150 **(b)** 180 **(c)** 240 **(d)** 120

9. The hour hand of a clock turns through how many degrees in 90 minutes? 1

(a) 46 **(b)** 50 **(c)** 44 **(d)** 45

10. 7 lines are drawn through a point O. The number of straight angles at O (pairs of opposite rays) is: 1

(a) 21 **(b)** 8 **(c)** 6 **(d)** 7

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. 10 rays start from O so that every pair makes an angle. Find the number of angles formed. 2

12. At what angle are the clock hands at 12:40? 2

13. A ray rotates 5 full turns and 270° more. Find the total angle turned and the number of right angles in it (nearest whole). 2

14. 12 rays start from O so that every pair makes an angle. Find the number of angles formed. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. At 9:50 how far (in degrees) is the minute hand ahead of the hour hand, and how far is it from 12? 3

16. Find the time after 8:00 when the minute hand has turned 60° and then state the smaller angle between hands. 3

17. A pizza is cut by 6 lines through the centre into equal slices (all lines passing through the centre). Find the angle of each slice and the number of slices. 3

OR

At 10:10 how far (in degrees) is the minute hand ahead of the hour hand, and how far is it from 12? 3

18. Find the time after 10:00 when the minute hand has turned 150° and then state the smaller angle between hands. 3

OR

A pizza is cut by 9 lines through the centre into equal slices (all lines passing through the centre). Find the angle of each slice and the number of slices. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. At 8:50, find 5
(a) the angle turned by the hour hand since 12 o'clock,
(b) the angle turned by the minute hand since 12,
(c) the smaller angle between the hands,
(d) its type.

OR

A robot starts facing North. It turns clockwise 225°, then 45° anticlockwise, then 180° clockwise. Find its final direction angle clockwise from North and the further turn needed to face North again. 5

20. At 11:10, find 5
(a) the angle turned by the hour hand since 12 o'clock,
(b) the angle turned by the minute hand since 12,
(c) the smaller angle between the hands,
(d) its type.

OR

A robot starts facing North. It turns clockwise 315°, then 135° anticlockwise, then 90° clockwise. Find its final direction angle clockwise from North and the further turn needed to face North again. 5

END OF QUESTION PAPER — Total: 40 Marks